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Master's degree in
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NATURAL LANGUAGE INFERENCE FOR BIOMEDICAL DOMAINS: ENHANCING ENTAILMENT CLASSIFICATION

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This thesis is not only the result of academic work but a reflection of the love, support, and belief I've received from the people who have stood beside me.

ABSTRACT

This thesis investigates biomedical entailment classification, a task within Natural Language Inference (NLI) that determines whether a clinical trial description supports or contradicts a given medical statement. The study uses the NLI4CT dataset, which pairs hypotheses with sections from breast cancer clinical trial reports. A combination of statistical baselines, prompting strategies, and lightweight fine-tuning methods are examined to identify effective approaches for domain-specific inference.
Qwen2-7B-Instruct was identified as the best-performing model across prompt-based evaluations. It was further adapted using parameter-efficient fine-tuning with LoRA and refined through a consistency-based postprocessing strategy. The final system achieved a macro F1-score of 0.78 on the test set, surpassing previous baselines. The results show that carefully selected model configurations and adaptation methods can improve performance in biomedical NLI tasks without requiring full-scale fine-tuning or extensive resources.

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1. INTRODUCTION

Is the hypothesis true? Natural Language Inference (NLI) addresses this by asking, for any given premise–hypothesis pair, whether the hypothesis is entailed by the premise, contradicted by it, or neither (neutral)[1].

NLI is a specialized task within the broader field of Natural Language Processing (NLP), falling under the umbrella of **textual entailment** and **semantic understanding**. Unlike simpler classification tasks, NLI requires models to grasp deeper relationships, synonymy, antonymy, causality, and real-world knowledge, ensuring that they can determine logical consistency rather than relying on surface-level patterns[2].

In general domain settings, NLI has played a key role in advancing representation learning and model design. Datasets like SNLI (Stanford Natural Language Inference) and MNLI (Multi-Genre Natural Language Inference)[3] have become standard benchmarks for evaluating how models interpret relationships between everyday sentences. While these benchmarks have supported major improvements in areas such as attention mechanisms and transfer learning[4], they mainly involve straightforward language and general scenarios.

In the biomedical domain, NLI is far more complex and calls for deeper investigation, clinical texts overflow with domain-specific terminology, overlapping abbreviations, and subtle statistical phrasing that can flip an inference from entailment to contradiction or demand a neutral verdict when an outcome isn't mentioned at all. Ensuring accurate, transparent judgments in this high-stakes setting requires dedicated methods and further research.

Recent studies in biomedical NLI have shown that model performance improves with well-designed prompting strategies. Instructional prompting structured few-shot templates, and light forms of chain-of-thought prompting have led to better inference accuracy by directing model attention to relevant clinical features [2], [5]. Other research has applied input rephrasing and ensemble prompt voting to reduce variation

in predictions and produce more consistent outputs in complex clinical scenarios[6].
This thesis carefully investigates different approaches to improve inference performance in clinical settings. By comparing multiple solutions and evaluating their impact, the work aims to identify effective ways of handling the complexity of biomedical NLI.

1.1 Objectives of the Study:

The primary goal of this thesis is to enhance performance in the **biomedical entailment classification task** by identifying effective combinations of model architectures, prompt designs, fine-tuning methods, and consistency strategies.
To achieve this goal, the following questions are addressed:

1. **Statistical vs. Large language Model Approach**
   How does the performance of the traditional statistical method compare to that of large language models in biomedical binary NLI classification tasks?
2. **Prompt Structure and Model Architecture**
   Which prompt structures are most effective across different LLM architectures?
3. **Few-Shot Learning**
   What is the impact of few-shot learning on model performance, and which model architecture benefits most from few-shot prompts in this setting?
4. **Fine-Tuning**
   Can lightweight fine-tuning methods improve the performance of the best-performing model?
5. **Consistency Strategies**
   How can we achieve stable and reliable predictions when applying inference models to biomedical NLI?

These questions guide the experimental design and define the contribution of this research to the broader effort of improving language inference in biomedical applications.

1.2 Significance and Scope of the Study:

Significance of the Study

In the biomedical field, Natural Language Inference (NLI) presents a significant challenge, as reading clinical trial reports requires both accuracy and complexity[2].

Natural Language Inference (NLI) presents persistent challenges in biomedical contexts, where understanding clinical trial reports demands both precision and domain-specific interpretation. Enhancing the ability to detect entailment and contradiction in this context can directly support evidence-based decision-making, streamline research analysis, and contribute to improved healthcare outcomes[7].

Recent advances suggest that model performance and output reliability benefit from deliberate prompt design, efficient fine-tuning, and mechanisms for reducing variability in predictions. These methods promote both effectiveness and stability, key requirements for applying automated systems in sensitive biomedical environments[6].

Despite progress in general-domain NLI, many models still struggle to perform reliably when applied to clinical texts, due to the complexity of biomedical language, limited annotated data, and the absence of domain-tuned inference strategies. Addressing this gap requires a careful combination of input formatting, tuning efficiency, and output consistency elements that this study integrates to explore scalable solutions within an existing clinical benchmark[5].

The broader significance of this work lies in its contribution to building robust and dependable automated systems for biomedical applications. As these systems are introduced into clinical and research workflows, producing consistent and trustworthy results becomes essential for gaining confidence and enabling responsible use.

Scope of the Study

This study focuses on **biomedical entailment classification**, specifically the binary classification of clinical statements as either entailed or

contradicted by descriptions from clinical trial reports. The dataset used throughout is the SemEval-2024 Task 2 (NLI4CT), ensuring consistency and comparability with prior research[8].

The work evaluates both traditional and modern language model-based approaches. A statistical baseline using TF-IDF for text representation is compared with pre-trained language models including Mistral, Qwen2, and Flan-T5. These models are assessed using zero-shot and two-shot prompting setups. Based on performance, the best model is selected for additional improvement using **LoRA-based fine-tuning**.

To reduce variability in predictions, the study applies **self-consistency techniques**, including output voting across repeated model runs. Evaluation is conducted using macro-averaged precision, recall, and F1 scores to measure both predictive performance and stability.

This study does not involve the development of new model architectures, dataset construction, or general-domain inference. Its scope remains strictly centered on improving the reliability and effectiveness of biomedical entailment classification using existing resources and methods.

1.3 Outline of the Thesis:

This thesis is organized into five chapters, each addressing a key component of the research process.

- **Chapter 1** introduces the motivation, objectives, significance, and scope of the study, providing a foundation for the work that follows.
- **Chapter 2** presents the background necessary to understand the task and its challenges. It includes an overview of Natural Language Inference (NLI), its role in biomedical applications, the development of language models, and a review of related literature.
- **Chapter 3** describes the materials and methods used in this study. It details the dataset, data preprocessing steps, model selection, and the experimental framework, including prompting strategies, fine-tuning with LoRA, and self-consistency methods.

- **Chapter 4** reports the results of the experiments, including data preprocessing steps and performance evaluations across different models and configurations.
- **Chapter 5** discusses the findings, compares them to related work, and highlights the implications, strengths, and limitations of the approach and concludes the thesis by summarizing the contributions and suggesting directions for future research.

2. BACKGROUND

This chapter provides the foundational knowledge necessary to understand the biomedical entailment classification task addressed in this thesis. It introduces Natural Language Inference (NLI), its applications in biomedical contexts, and the evolution of modeling techniques used to solve NLI tasks. The chapter also includes a review of key literature and highlights how recent developments have shaped current challenges and opportunities in this field.

2.1 From Natural Language Processing to Natural Language Inference:

Over the past decade, Natural Language Processing has evolved into one of the most impactful fields in artificial intelligence. Its purpose is to allow machines to comprehend, interpret, and produce human language in meaningful and effective ways. From voice assistants and machine translation to search engines and text summarization, NLP has become deeply integrated into everyday life [2].

NLP combines linguistics, computer science, and machine learning to allow computers to process large amounts of text and extract useful information. Among the many tasks NLP covers, such as entity recognition, sentiment analysis, and machine translation, one essential and challenging task is Natural Language Inference (NLI). NLI is widely regarded as a core benchmark for evaluating a system's capacity for language understanding, as it requires a model not only to process information but also to draw logical conclusions from it [1].

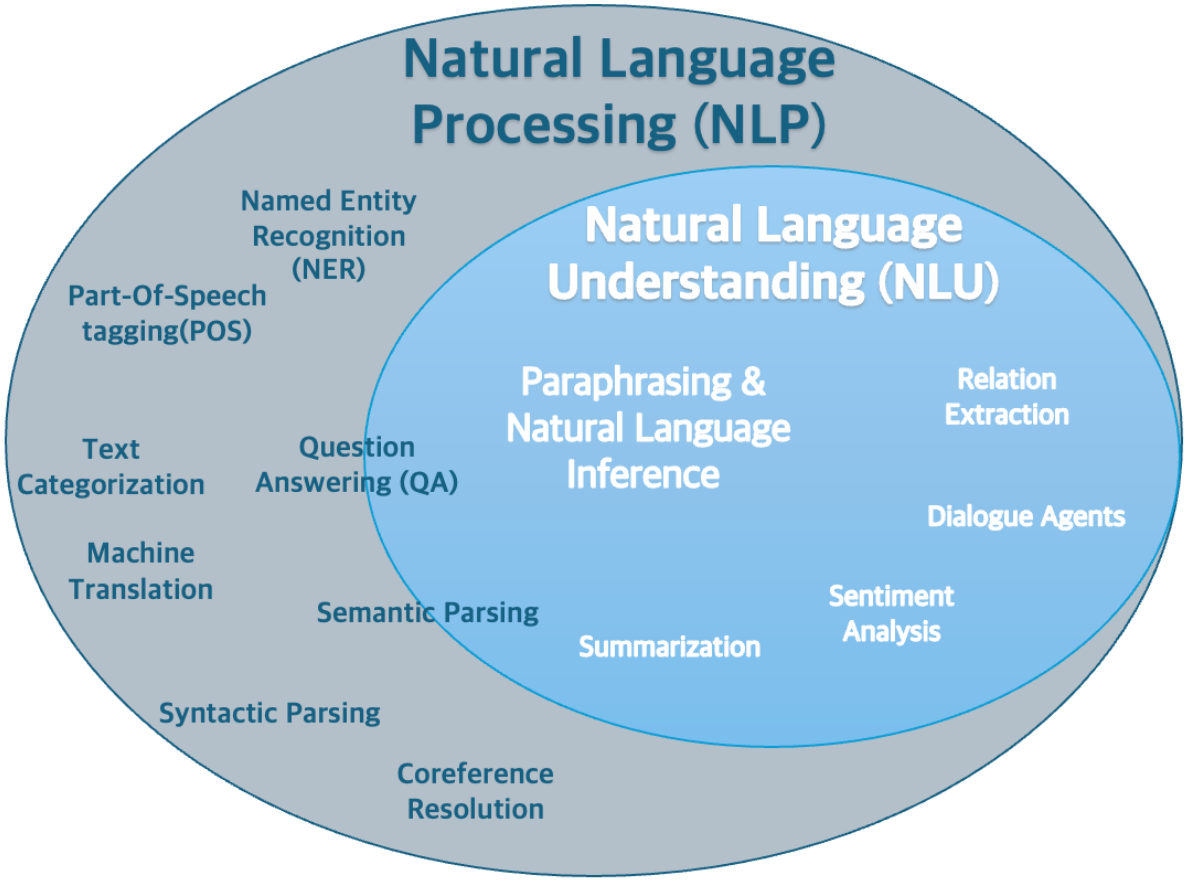

Figure 2.1 –Structural overview of NLP and its subfields.

Natural Language Inference, sometimes called textual entailment, is concerned with determining whether a specific statement (hypothesis) logically follows from another statement (premise). It may sound simple, but this task requires a deep understanding of language, logic, and context. To illustrate, consider the following:
Premise: "All participants in the study received a flu vaccine."
Hypothesis: "The study included patients who were vaccinated."
The hypothesis is clearly supported by the premise here. This is an example of **entailment**, a core category in NLI.

NLI typically classifies such relationships into three categories:

- **Entailment**: The hypothesis must be true if the premise is true.
- **Contradiction**: The hypothesis directly conflicts with the premise.

- **Neutral**: The hypothesis could be true or false; the premise doesn't provide enough information [1].

While NLI is already difficult in general-purpose texts, applying it to biomedical documents, such as clinical trial reports, introduces even greater complexity. The language is more technical, the phrasing more compact, and the reasoning often depends on domain-specific knowledge [2].

This thesis focuses on advancing NLI in such biomedical contexts, where making the proper inference could impact research outcomes or clinical decisions. In the following section, we will explore these biomedical-specific challenges in more detail.

2.2 Natural Language Inference in Biomedical Applications:

Natural Language Inference, widely accepted as a benchmark in open-domain natural language processing, faces distinct challenges when applied to biomedical texts. Clinical trial reports, clinical notes, and scientific abstracts often contain specialized terminology, domain-specific jargon, and complex sentence structures. Unlike general language, biomedical literature includes intricate causal relationships, implicit assumptions, and subtle distinctions that require expert interpretation [9].

In this context, models are tasked with determining whether a given statement, such as the outcome of a clinical claim or a diagnostic result, can be inferred from a relevant source document. This is a technically demanding and high-stakes task, as such inferences can inform medical decision-making and support evidence synthesis in research[10].

To meet the need for domain-adapted evaluation, several biomedical NLI datasets have been introduced. One early example is MedNLI, a dataset built from clinician-annotated clinical notes, designed to reflect the reasoning challenges encountered in practice. More recently, the NLI4CT dataset, released through the SemEval-2024 Task 2 competition, focuses

specifically on inferring whether a statement about a clinical trial is supported or contradicted by the trial report [8].

While large language models have shown strong results in general-domain NLI, their performance tends to degrade in biomedical contexts. Factors such as vocabulary mismatch, limited domain adaptation, and reliance on surface-level patterns can reduce reliability. Addressing these issues requires domain-specific training and strategies that improve model robustness, consistency, and task performance [9].

To address these challenges, the current work investigates how different model architectures, prompt styles, and fine-tuning strategies can be optimized for entailment classification in biomedical domain tasks. The following section reviews the development of language models that support this work.

2.3 Development of Language Models:

Progress in Natural Language Processing and Inference has closely followed the evolution of language modeling. Since 2018, major advances in pretraining techniques and transformer-based architectures have improved performance across both general and specialized tasks, including biomedical inference.

Language modeling has shifted from basic statistical techniques to more adaptable systems capable of handling varied text patterns. This transition has supported stronger generalization and better use of existing data, making it possible to apply language models in complex domains with limited supervision [4].

The next sections explain this progression by first outlining statistical methods and then describing the rise of large language models.

2.3.1 Statistical Approaches in NLI:

Earlier developments in language modeling were grounded in statistical methods that used frequency-based patterns to represent and interpret text. These approaches relied on the idea that word usage across large corpora could offer useful signals for language understanding.

Among the common techniques were n-gram models, which used short word sequences to estimate how likely one word is to follow another, and bag-of-words, where the order of words is ignored and text is treated as a set of individual terms. To compare or evaluate texts, these models often applied measures such as TF-IDF, which stands for term frequency–inverse document frequency. TF-IDF scores each word based on how often it appears in a specific document (term frequency) and how rare it is across all documents (inverse document frequency). Words that are frequent in a document but uncommon in the overall corpus are given higher importance, allowing the model to focus on terms that are more distinctive [11].

Another widely used technique was cosine similarity, which measures the angle between two vectors representing text documents. After converting texts into vectors using methods like bag-of-words or TF-IDF, cosine similarity calculates how closely those vectors point in the same direction. A score of 1 means the texts are identical in terms of word composition, while 0 indicates no overlap. This allows the model to assess textual similarity regardless of length [12].

These methods supported early work in classification, retrieval, and basic semantic comparison.

When applied to Natural Language Inference, statistical models compared the presence or absence of overlapping terms between the premise and hypothesis. Their predictions were typically based on shallow similarity, making them suitable only for straightforward cases.

In biomedical tasks, where language tends to be more technical and precise, the limits of these models become more evident. Even so, they serve as a practical baseline to contrast the benefits of more advanced architectures.

2.3.2 Evolution of Large Language Models (LLMs):

The shift from statistical methods to neural-based models transformed the field of language modeling. Instead of depending on word frequency patterns, modern models learn contextual relationships from large-scale data. Early developments like word embeddings laid the foundation by

representing words in dense vector spaces, enabling models to capture semantic similarity.

The introduction of transformer-based architectures, such as BERT[4] and GPT[13], marked a major leap forward. These models use self-attention mechanisms to understand dependencies between words, regardless of their position, allowing for more flexible and accurate language comprehension. With scale, these architectures evolved into large language models (LLMs), trained on massive corpora and capable of general-purpose reasoning and generation. A timeline of major developments in LLMs is illustrated in Figure 2.2 to highlight the key architectural and capability milestones leading up to recent models.

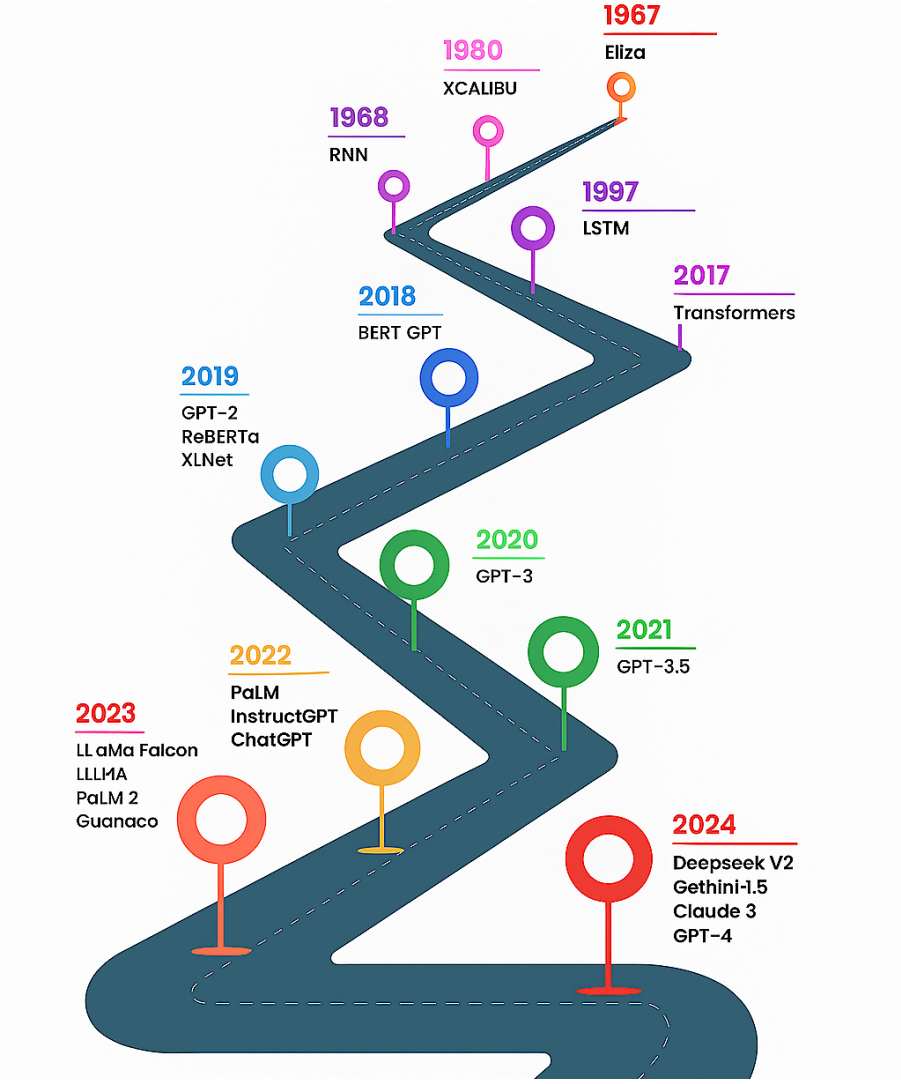

Figure 2.2 – Timeline of large language model development (1967–2024)

LLMs can perform a wide range of language tasks, including summarization, translation, question answering, and inference, without task-specific tuning. Their strength lies in their ability to generalize from limited examples and adapt to new domains with minimal modification [14].

Among the models used in this work:

- Flan-T5 is an encoder-decoder architecture built on the T5 framework, further trained with instruction tuning. It processes the input sequence through an encoder and generates the output via a separate decoder, making it well-suited for tasks involving transformation or structured generation. Its training objective is to make the model better at following natural language instructions across a wide set of tasks [15].
- Mistral follows a decoder-only transformer architecture with sliding window attention and grouped-query attention, which improves efficiency for long contexts without sacrificing performance. It is optimized for fast generation while maintaining strong performance on reasoning and classification benchmarks [16].
- Qwen2 is also a decoder-only transformer, trained on a large mixture of text and code in multiple languages. It emphasizes multi-task instruction tuning and few-shot generalization, making it a strong candidate for tasks that require adaptability and precise output formatting [17].

While these models show impressive performance on general-domain tasks, biomedical entailment classification presents distinct challenges due to domain-specific language, implicit logic, and subtle contradictions. To address this, further adaptation through prompting, few-shot examples, and fine-tuning techniques is often necessary.

2.4 Literature Review:

Research on Natural Language Inference (NLI) has evolved from statistical models to transformer-based architectures, primarily driven by

advances in large-scale language modeling. In open-domain NLP, benchmarks such as SNLI [1] and MNLI [3] have played a central role in training and evaluating models across diverse inference scenarios. While these datasets contributed significantly to representation learning and transferability, their applicability in biomedical NLI remains limited. This is due to their lack of domain-specific terminology, absence of structured biomedical relationships, and limited representation of implicit clinical reasoning patterns. For instance, unlike SNLI, biomedical texts often rely on statistical claims, shorthand for drug regimens, or contextual inference over multi-sentence passages, none of which are present in general-domain corpora.

To bridge this gap, specialized resources have been introduced. MedNLI [2] offers a clinically annotated dataset derived from physician notes, targeting reasoning challenges in medical settings. More recently, NLI4CT [7][8] was introduced through SemEval-2024 Task 2 to evaluate whether a clinical trial report supports, contradicts, or does not address a given hypothesis. This dataset focuses on multi-evidence judgment grounded in trial outcomes, making it more aligned with real-world biomedical reasoning tasks.

Recent research has emphasized the importance of prompt design in adapting language models to specialized domains. Prompting techniques guide the model to interpret tasks more effectively by shaping its contextual understanding. Studies such as Schick & Schütze (2021) on Pattern-Exploiting Training (PET) and domain-tuned models like BioPromptBERT have shown that structured prompt templates can significantly improve performance by embedding task-relevant patterns [6][12]. Instruction-based prompting and refined few-shot templates have also been shown to direct model attention toward salient biomedical concepts, improving both performance and consistency.

Nevertheless, deploying large language models in biomedical NLI remains challenging due to issues like vocabulary mismatch, limited supervision from domain-specific data, and inconsistencies in prediction outputs. To overcome these limitations, this thesis introduces a combined approach that benchmarks instructional prompting under few-shot settings, applies lightweight fine-tuning techniques such as parameter-

efficient adaptation, and evaluates model stability through output voting. These techniques are explored to improve performance and consistency in entailment classification for biomedical domain tasks.

3. MATERIALS & METHODS

This chapter walks through the full pipeline developed to tackle entailment classification in biomedical contexts. After preparing the data, the process unfolds in several focused stages, each one aimed at examining, testing, and improving model performance while keeping computational demands in check.

As illustrated in Figure 3.1, everything starts with a simple statistical baseline using TF-IDF, giving us a clear point of comparison. We then move into prompt engineering, where different instructional formats are tested across three language models, Flan-T5, Mistral, and Qwen2. By experimenting with multiple ways of structuring the prompts, we aim to help each model better understand and handle the task. Based on their overall performance, including results from both zero-shot and few-shot settings, we identify the most promising combination of model and prompt strategy for further optimization. This selected model is then fine-tuned using a lightweight approach (LoRA), allowing domain adaptation without requiring large-scale computational resources. To improve output stability, especially in borderline cases, a self-consistency mechanism is applied during inference by aggregating multiple predictions through majority voting. The full pipeline supports reproducibility, model efficiency, and performance gains, while remaining feasible within limited-resource academic environments.

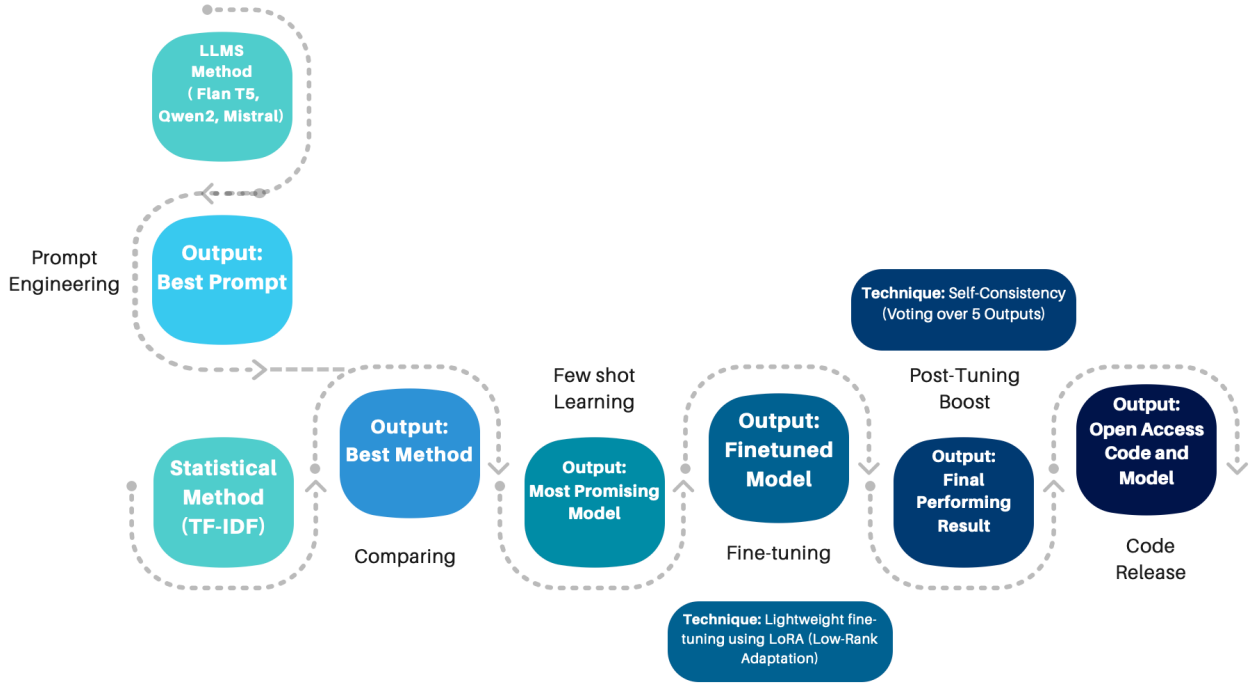

Figure 3.1 – Overview of the methodological pipeline

3.1 Dataset Description:

The dataset used in this study is the Natural Language Inference for Clinical Trials (NLI4CT) dataset, introduced as part of the SemEval-2023 and SemEval-2024 shared tasks [7], [8]. It contains 2,400 expert-annotated examples drawn from breast cancer clinical trial reports (CTRs) hosted on ClinicalTrials.gov. The dataset is grounded in real-world biomedical texts and was constructed to facilitate systematic evaluation of natural language inference systems in specialized domains.

Annotations were provided by clinical domain experts, including clinical trial organizers and research oncologists from the Cancer Research UK Manchester Institute and the Digital Experimental Cancer Medicine Team, ensuring medically grounded and reliable labels.

Each data point includes a hypothesis, one or two clinical trial identifiers (IDs), and a section name, which indicates which part of the trials will be used to extract the premises, with the relationship labeled as:

- **Entailment**: The premise supports the hypothesis
- **Contradiction**: The premise refutes the hypothesis

All premises are extracted from a single section of the CTR, either eligibility criteria, interventions, results, or adverse events. If two CTRs are present, both are drawn from the same section to preserve contextual alignment.

These sections represent standardized components of clinical trial reports:

- **Eligibility criteria** outline the medical and demographic conditions required for patients to participate in the trial.
- **Intervention** describes the treatment protocols under investigation, including dosage, frequency, and duration.
- **Results** summarize clinical outcomes, including participant numbers, success measures, and follow-up timelines.
- **Adverse events** report any side effects or complications observed during the trial.

Figure 3.2 below visually summarizes the feature extraction process. Starting from the original dataset, which contains trial IDs, section names, labels, and hypotheses, the relevant section(s) are retrieved from the corresponding clinical trial(s). These extracted sections are then used to construct the premise(s), forming the final input structure used for entailment classification.

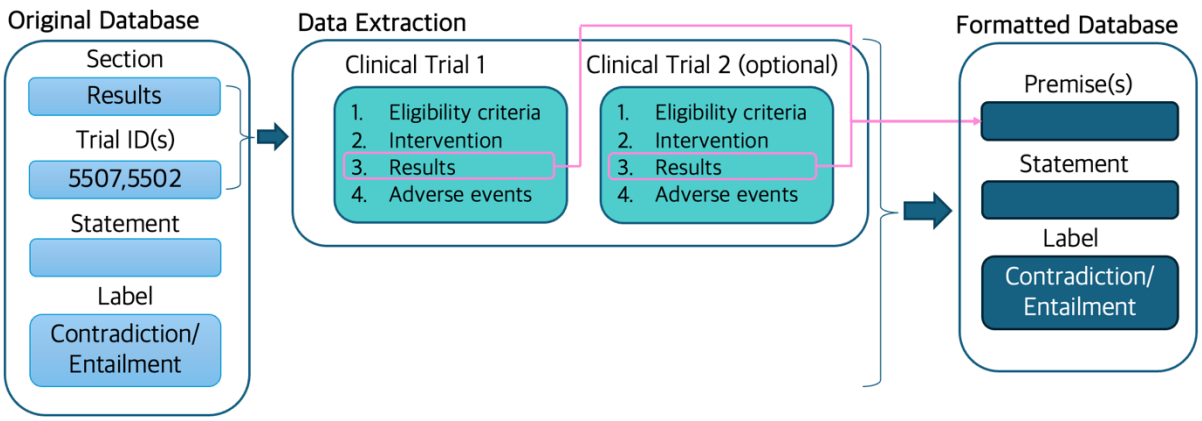

Figure 3.2 – Data preparation process

Figure 3.3 illustrates a labeled data point from the dataset used for entailment classification. Each instance includes a hypothesis (referred to as the statement), one or two premise segments derived from clinical trial reports, and a corresponding label indicating whether the premise(s) support or contradict the hypothesis. In this example, both the primary and secondary trials involve MRI procedures, supporting the hypothesis and thus labeled as "Entailment." This structure reflects the real-world clinical inference task where multiple trials may provide evidence for a single claim, requiring precise extraction and interpretation of domain-specific content.

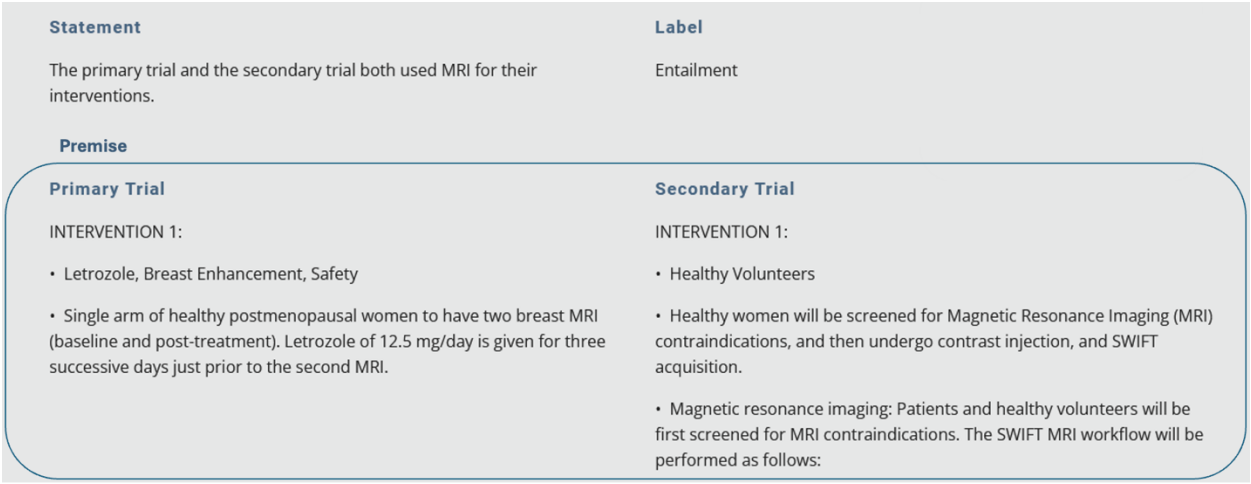

Figure 3.3 – Example of a labeled data point used for entailment classification.

The dataset is divided into:

- **Training** set: 1,700 samples
- **Development** set: 200 samples
- **Test** set: 500 samples

This structure supports targeted evaluation and experimentation, particularly in tasks involving entailment classification in biomedical domain settings.

3.2 Methodological Framework:

This section outlines the step-by-step strategy implemented to improve performance in entailment classification within biomedical domain tasks. The methodology spans from traditional statistical modeling to large language model-based approaches, incorporating few-shot prompting, lightweight fine-tuning, and post-tuning consistency mechanisms.

3.2.1 Statistical Approach (TF-IDF):

To establish a transparent baseline for entailment classification, a statistical method using Term Frequency–Inverse Document Frequency (TF-IDF) combined with cosine distance was implemented. This approach transforms both the hypothesis and clinical trial premise(s) into weighted vector representations, then evaluates their semantic similarity based on vector alignment.

TF-IDF scores are computed by multiplying term frequency with an inverse document frequency adjustment:

$$\boldsymbol{TF - IDF(t, d, D) = TF(t, d) \times log\left(\frac{N}{DF(t)}\right)}$$

where:

$TF(t, d)$ is how often term t appears in document d,

$DF(t)$ is the number of documents containing term t,

N is the total number of documents in the corpus.

In this context, a "document" refers to each individual sentence within the selected clinical trial section. When computing TF-IDF, all entries from that section are treated as the corpus. For example, if the Eligibility section contains ten lines, those ten lines constitute ten documents. The document frequency (DF) of a term is then the number of those lines that include the term. This localized corpus allows the TF-IDF scores to reflect the relative importance of terms within a specific trial context.

To illustrate, consider the following pair:

Hypothesis: "The study included women over 65 years old."

Premise (Eligibility Section): "Women aged 65 and older were eligible to participate."

The TF-IDF vectors constructed from these texts assign higher weights to informative terms such as “women,” “65,” and “eligible,” while discounting common words. Cosine distance between the hypothesis vector H and the premise vector P is calculated as:

$$CosineDistance(H, P) = 1 - \frac{(H \cdot P)}{(\|H\| \cdot \|P\|)}$$

For this example, the resulting cosine distance was 0.27, indicating strong semantic alignment between the hypothesis and premise.

Based on such observations across the development set, we established a decision threshold of 0.9:

If the average cosine distance > 0.9 → Contradiction

If the cosine distance ≤ 0.9 → Entailment

This rule was consistently applied across single-premise and dual-premise scenarios. When two clinical trials were present, cosine distances from both were computed and averaged before classification.

3.2.2 Model Selection:

This study evaluates three instruction-tuned language models for the entailment classification task: Flan-T5 XL, Mistral-7B-Instruct, and Qwen2-7B-Instruct. Each model represents a different architecture and training setup, allowing a focused comparison on performance within the biomedical domain.

Flan-T5 XL is an encoder-decoder model based on the T5 architecture, with around 3 billion parameters. Its structure separates input encoding from output generation, making it well-suited for classification tasks. It has been instruction-tuned across a wide range of datasets, which improves its response to structured prompts and helps it generalize across various NLP tasks [6].

Mistral-7B-Instruct is a decoder-only transformer with 7 billion parameters. It includes grouped-query attention and sliding-window attention to handle longer sequences efficiently. The model is optimized for instruction-following and has shown strong results in reasoning and classification benchmarks, including general NLI tasks [16].

Qwen2-7B-Instruct, developed by Alibaba Cloud, is also a decoder-only model with 7 billion parameters. It is trained on instruction-following tasks and supports long-context processing. The model performs well on complex classification prompts and has demonstrated strong stability in output generation [17].

These models were selected to compare different transformer architectures (encoder-decoder vs. decoder-only) and to examine their performance without domain-specific pretraining. Each is tested under few-shot settings using a unified prompt format. The goal is to assess how well they support biomedical entailment classification when combined with lightweight fine-tuning and post-processing strategies for prediction stability.

3.2.2 Zero- & Few-Shot Learning Approaches:

We began by testing each model under zero-shot settings, where only an instructional prompt was provided. The models were asked to classify whether a clinical statement was entailed or contradicted by the given trial text. Zero-shot results were unstable and often misaligned with the task requirements.

To improve performance, we evaluated over ten prompt variations per model. These included changes in wording, structure, and the position of the hypothesis within the instruction. One notable improvement was observed when the statement was integrated directly into the question prompt rather than being presented as a separate input. This adjustment helped the model focus more effectively on the intended inference task.

One-shot prompting was avoided. Early tests showed that a single example frequently caused output bias toward the demonstrated class. We adopted two-shot prompting instead, with each prompt containing one entailment and one contradiction example. The same pair of examples was used consistently across all evaluations to prevent data leakage. Two-shot prompting provided a stronger contextual grounding for the task and reduced class imbalance in model outputs, helping stabilize predictions across both known and difficult cases.

This setup enabled a fair comparison across models in low-resource conditions prior to fine-tuning. The exact prompt formats used in both zero-shot and two-shot settings are provided in Appendix A.

3.2.2 Fine-Tuning:

From the previous step, the most promising model, with the strongest performance across prompting configurations, was selected for fine-tuning.

Adapting language models to domain-specific tasks such as biomedical entailment classification typically benefits from fine-tuning on curated task-relevant data. Full-model fine-tuning, however, requires substantial computational resources and time due to the size of modern transformer-based architectures. To address these constraints, we opted for a lightweight fine-tuning strategy using Low-Rank Adaptation (LoRA).

Several parameter-efficient fine-tuning (PEFT) methods have been proposed to reduce the cost of model adaptation while maintaining performance, including Adapters, Prefix Tuning, and LoRA. Among these, LoRA introduces trainable low-rank matrices into the attention mechanisms of the transformer architecture, allowing task-specific knowledge to be integrated without updating the full parameter space.

We selected LoRA due to its balance between efficiency and effectiveness. Unlike full fine-tuning, which modifies all weights in the model, LoRA freezes the original weights and only updates a small number of additional parameters. This significantly reduces memory usage and training time, making it suitable for research settings with limited computational resources.

The fine-tuning process was applied to the instruction-formatted version of the NLI4CT training set, comprising 1,700 examples. Each prompt presented the trial evidence and hypothesis in a natural language format, followed by a classification request. LoRA modules were inserted into the self-attention layers of the model, allowing it to learn domain-specific patterns.

Training was conducted using a batch size of 8, a learning rate of 2e-5, and a linear scheduler with warm-up. The model was trained for three epochs, with validation on a 200-example development set. The best-performing checkpoint based on macro F1-score was retained for evaluation.

This approach provided a practical and scalable way to tailor large language models to biomedical inference tasks while preserving model integrity and limiting resource consumption.

3.2.3 Self-Consistent Learning:

Despite improvements achieved through fine-tuning, the model occasionally produced unstable or inconsistent predictions, particularly on borderline cases with overlapping or ambiguous clinical statements. To address this, we implemented a self-consistency strategy aimed at stabilizing output decisions through repeated inference.

Self-consistent learning involves generating multiple outputs for the same input using randomized decoding (e.g., sampling with temperature) and aggregating the responses through majority voting. This method is based on the observation that while single outputs may vary, the most frequent prediction across several runs often reflects the most robust interpretation.

In our setup, each test example was passed through the fine-tuned model multiple times (five) using a fixed temperature and random seed variation. The model's final label prediction was determined by majority vote over these multiple outputs.

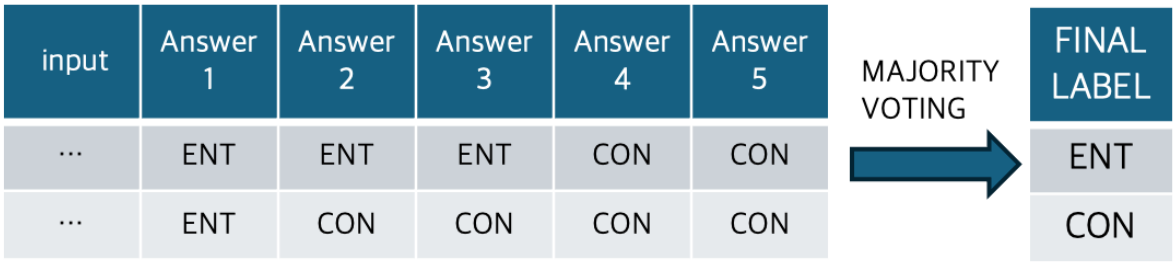

Figure 3.4 – Majority voting scheme used in self-consistent inference. The final label is chosen based on the most frequent prediction across multiple outputs for each input.

This technique was applied only during inference and required no additional training. It offered a lightweight mechanism to enhance output reliability and reduce prediction variance without modifying the model parameters.

3.3 Evaluation Metrics:

performance on the entailment classification task, we used **precision**, **recall**, and **F1-score**, calculated under a **macro-averaging scheme**. These metrics assess how well the model distinguishes between the two labels: **Entailment** and **Contradiction**.

In classification tasks, two common averaging methods are used:

Micro-averaging computes global metrics by aggregating all true positives, false positives, and false negatives across all classes. It is influenced by class distribution and is suitable when classes are imbalanced and the overall number of predictions is the focus.

Macro-averaging computes metrics independently for each class and then averages them. This method gives **equal importance to each class**, regardless of how many examples it contains.

Given that our dataset has a **balanced class distribution** and the goal is to **treat Entailment and Contradiction equally**, macro-averaging was preferred. It ensures that both classes contribute equally to the evaluation, avoiding bias toward the more frequently predicted class.

Let TP = true positives, FP = false positives, and FN = false negatives for each class.

Macro Precision:

$$\text{Precision}_{macro} = \frac{1}{2}\left(\frac{TP_{entail}}{TP_{entail} + FP_{entail}} + \frac{TP_{contra}}{TP_{contra} + FP_{contra}}\right)$$

Macro Recall:

$$\text{Recall}_{macro} = \frac{1}{2}\left(\frac{TP_{entail}}{TP_{entail} + FN_{entail}} + \frac{TP_{contra}}{TP_{contra} + FN_{contra}}\right)$$

Macro F1-Score:

$$\text{F1}_{macro} = \frac{1}{2}(F1_{entail} + F1_{contra})$$

where for each class:

$$F1 = 2\,\frac{Precision \cdot Recall}{Precision + Recall}$$

All evaluation metrics were computed on the 500-sample test set. Predictions from each model configuration, whether derived from prompt-based inference, fine-tuned models, or post-processed outputs, were compared against the corresponding ground truth labels. Macro F1-score was used as the primary metric for comparison, as it provides equal weight to both entailment and contradiction classes, making it more suitable for evaluating balanced performance across labels.

4. RESULTS

This chapter reports the results of all evaluated model configurations for the binary entailment classification task. Performance is compared across five stages: statistical baseline, zero-shot prompting, few-shot prompting, lightweight fine-tuning, and self-consistency. Metrics include macro precision, macro recall, and macro F1-score, computed on the 500-example test set. Each section answers one of the research questions by examining how different techniques influence model effectiveness.

4.1 Model Results:

4.1.1 Statistical vs. Large Language Model Performance:

This section compares the performance of the statistical baseline with that of large language models on the biomedical entailment classification task. The TF-IDF model, which relies on lexical similarity using cosine distance, reached a macro F1-score of 0.46. Large language models outperformed this baseline in zero-shot settings. Flan-T5-XL reached 0.57, Mistral-7B achieved 0.60, and Qwen2-7B obtained the highest score with 0.66. These results correspond to the best prompts identified during prompt engineering. Table 4.1 summarizes these comparisons, showing that models capable of interpreting clinical language and structure achieved better alignment with the task requirements than the statistical method.

MODEL	MACRO PRECISION	MACRO RECALL	MACRO F1
TF-IDF BASELINE	0.46	0.46	0.46
FLAN-T5-XL ZERO-SHOT	0.57	0.57	0.57
MISTRAL-7B ZERO-SHOT	0.61	0.61	0.60
QWEN2-7B ZERO-SHOT	0.66	0.66	0.66

Table 4.1 – Zero-shot performance comparison between statistical method and LLMs

4.1.2 Effect of Prompting Strategies on Model Performance:

This section compares how different models respond to changes in prompt structure. In the zero-shot setting, we tested over ten different prompt variations per model, refining the format to improve clarity and alignment with the task. These prompt engineering efforts led to a notable improvement of approximately five macro F1-score points from the initial trials to the final selected prompts.

The most effective format involved directly integrating the hypothesis into the question. This structure consistently improved model behavior across all three architectures. For the two-shot setting, we reused this same prompt format and simply added two representative examples before the test input to guide the model.

We evaluated Flan-T5-XL, Mistral-7B, and Qwen2-7B under both settings. As shown in Figure 4.1, two-shot prompting further improved performance: Flan-T5-XL increased from 0.57 to 0.60, Mistral-7B from 0.60 to 0.62, and Qwen2-7B from 0.66 to 0.69. These results confirm that incorporating structured examples enhances model understanding in entailment classification.

Qwen2-7B achieved the highest scores in both configurations and demonstrated the most stable gains. Based on this, we selected it for further fine-tuning.

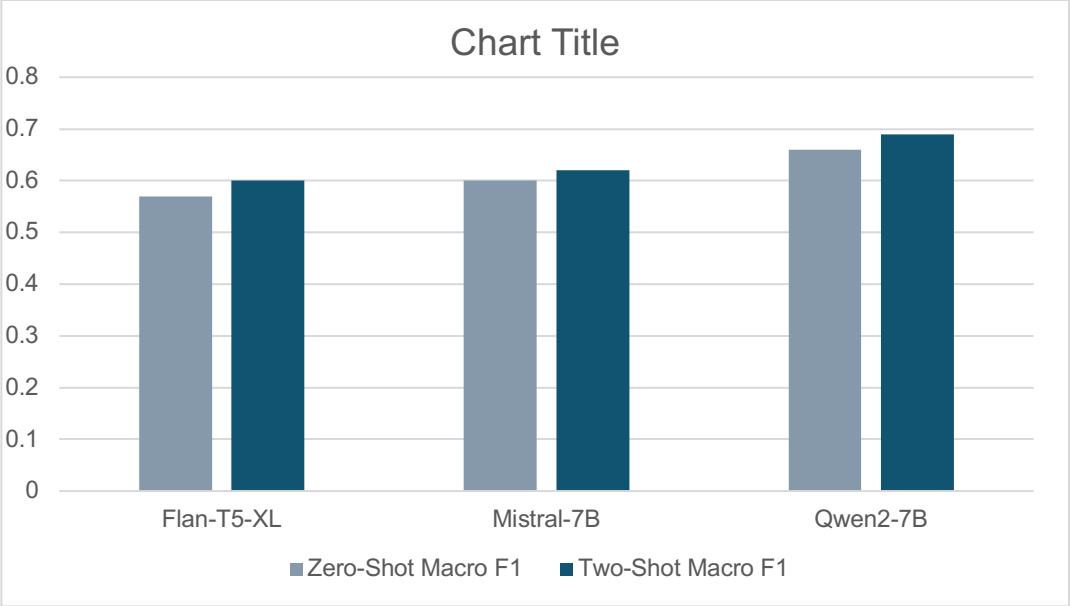

Figure 4.1 – Comparison of macro F1-scores across prompt settings for each model

4.1.3 Fine-Tuning Results:

Following the selection of Qwen2-7B-Instruct as the most promising model, we applied lightweight fine-tuning using LoRA to improve performance on the biomedical entailment task. The results are summarized in Table 4.2.

Prior to fine-tuning, the model achieved a macro F1-score of 0.66 under zero-shot prompting. After fine-tuning, this increased to 0.74, marking a notable improvement. Macro precision and recall also rose from 0.66 to 0.75 and 0.74, respectively. These gains show that even without full parameter updates, adapting the model to task-specific data enhanced its ability to distinguish between entailment and contradiction cases.

MODEL	MACRO PRECISION	MACRO RECALL	MACRO F1
QWEN2-7B ZERO-SHOT	0.66	0.66	0.66
QWEN2-7B ZERO-SHOT-FINETUNED	0.75	0.74	0.74

Table 4.2 – Comparison of Qwen2 performance before and after lightweight fine-tuning.

4.1.4 Post-Fine-Tuning Boost with Self-Consistency:

To improve prediction stability after fine-tuning, we applied a post-fine-tuning boost using self-consistency. The same input was passed to the model multiple times, and predictions were aggregated through majority voting. Only odd numbers of generations were tested to avoid tie outcomes. We evaluated 3, 5, and 7 generations; the best results were obtained using 5 outputs.

The base fine-tuned model achieved a macro F1-score of 0.74. With self-consistency (5 generations), the score improved to 0.78. Macro precision increased from 0.75 to 0.79, and macro recall from 0.74 to 0.78.

MODEL	MACRO PRECISION	MACRO RECALL	MACRO F1
QWEN2-7B ZERO-SHOT-FINETUNED	0.75	0.74	0.74
QWEN2_7B_ZERO_SHOT_FINETUNED (SELF-CONSISTENCY=5)	0.79	0.78	0.78

Table 4.3 – Performance improvement after applying self-consistency to the fine-tuned model.

4.2 Summary:

This section summarizes how the performance of the Qwen2-7B-Instruct model changed as different techniques were applied. The goal was to track improvements in macro F1-score across stages: prompting, fine-tuning, and output voting.

We also compare its performance with other models (Flan-T5-XL and Mistral-7B) in both zero-shot and two-shot settings, along with a statistical TF-IDF baseline. The table below presents macro precision, recall, and F1-scores for each configuration tested:

MODEL	MACRO PRECISION	MACRO RECALL	MACRO F1
TF_IDF	0.46	0.46	0.46
FLAN_T5_XL_ZERO_SHOT	0.57	0.57	0.57
MISTRAL-7B-ZERO-SHOT	0.61	0.61	0.60
QWEN2_7B_ZERO_SHOT	0.66	0.66	0.66
FLAN_T5_XL_TWO_SHOT	0.61	0.60	0.60
MISTRAL-7B-TWO-SHOT	0.64	0.62	0.62
QWEN2_7B_TWO_SHOT	0.69	0.68	0.69
QWEN2_7B_ZERO_SHOT_FINETUNED	0.75	0.74	0.74
QWEN2_7B_ZERO_SHOT_FINETUNED (SELF CONSISTENCY=5)	0.79	0.78	0.78

Table 4.4 – Model Performance Across Settings

The model was first tested using a zero-shot prompt. Then, two-shot prompting was introduced, which included one entailment and one contradiction example. The best-performing prompt from this stage was selected for the next step. Using that prompt format, the model was fine-tuned with LoRA, a lightweight approach that updates only a small set of parameters. Finally, we applied self-consistency by generating multiple outputs per input and using majority voting to decide the final prediction. We tested different odd numbers of generations (3, 5, and 7); the setting with 5 generations gave the best results.

The bar chart below shows the macro F1-scores from each step:

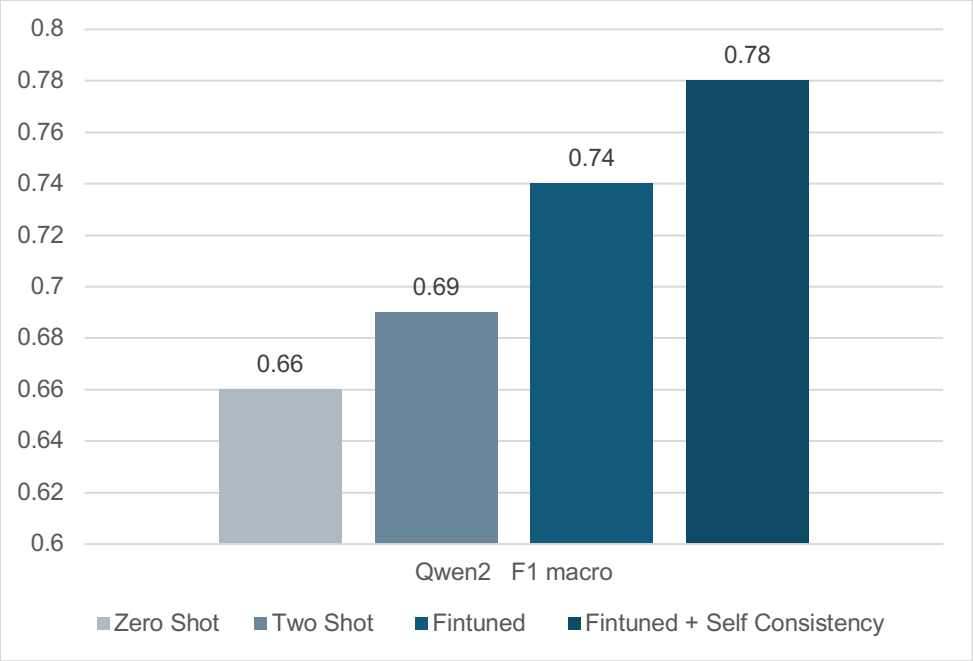

Figure 4.2 – Macro F1-Score Improvement of Qwen2-7B Across Methods

Each step brought a measurable improvement in classification performance. The final configuration, fine-tuned model with self-consistent output, produced the highest F1-score on the test set.

5. DISCUSSION & CONCLUSION

This chapter reviews the outcomes of the experiments, connects them to the research questions, and outlines the key observations, challenges, and potential directions for future work. It provides a complete view of how the methods contributed to the task of entailment classification in the biomedical domain.

5.1 Key Findings:

The evaluation showed that traditional statistical models like TF-IDF with cosine distance lack the ability to understand the semantic complexity found in clinical texts. Its macro F1-score of 0.46 confirmed that frequency-based representations are not sufficient when the task involves clinical reasoning, temporal conditions, or subtle phrasing variations.

Among the three language models evaluated in zero-shot settings, Qwen2-7B-Instruct delivered the strongest performance, achieving a macro F1-score of 0.66, ahead of Mistral-7B and Flan-T5 XL. This result suggests that Qwen2's training and architecture are better aligned with the type of inferential answering required in clinical trial texts. Improvements were observed across all models when using two-shot prompting. The consistent format and balanced class examples helped the models better focus on the task and reduced class bias in their outputs.

After identifying Qwen2 as the most promising model, it was fine-tuned using LoRA. This approach allowed the model to adjust its predictions based on domain-specific training data while keeping the full parameter space intact. The fine-tuned model reached a macro F1-score of 0.74, showing meaningful gains over zero-shot and few-shot setups.

To reduce variance in predictions, a self-consistency method was applied by generating multiple outputs per instance and selecting the most

frequent prediction. Using a voting count of five, this method increased Qwen2's final performance to a macro F1-score of 0.78. This step addressed inconsistencies observed in some borderline cases and stabilized model outputs.

5.2 Related Work:

Recent contributions from the SemEval-2024 shared task provide a benchmark for biomedical entailment classification on the NLI4CT dataset. Two of the top-performing systems, GU-NLP and TaNLP, demonstrated effective strategies that combined prompt engineering and model adaptation techniques.

The GU-NLP team employed a multi-stage pipeline combining BM25-based sentence selection—a traditional term-frequency ranking method—and MPNet reranking, a transformer-based model that refines relevance based on semantic similarity. This approach aimed to reduce input size while preserving contextual relevance. They then applied instruction-based prompting and an ensemble method to generate stable predictions. This system achieved a macro F1-score of 0.745 on the test set[18].

The TaNLP team combined Flan-T5 XXL with DeBERTa-v3 large and applied LoRA-based fine-tuning. Their approach introduced contrastive hypothesis pairs to improve classification boundaries and incorporated a lightweight ensemble of classifiers. Their final system achieved a macro F1-score of 0.754[19].

In our study, we followed a more streamlined approach. After evaluating several models in zero-shot and two-shot settings, Qwen2-7B-Instruct emerged as the most promising. We fine-tuned it using LoRA on instruction-formatted NLI4CT data and applied a self-consistency method to refine predictions. This setup, without ensemble techniques or external reranking, achieved a macro F1-score of 0.78.

Compared to GU-NLP and TaNLP, our method avoided multi-stage pipelines and still outperformed both. This suggests that well-structured

prompts, lightweight fine-tuning, and post-processing can offer strong results without high computational overhead.

5.3 Answers to Research Questions:

Main Research Question:

What combination of modeling architecture, prompt designs, and fine-tuning techniques and consistency strategies most effectively enhances performance in biomedical entailment classification tasks?

To answer this, five sub-questions were examined:

1. **How does the performance of the traditional statistical method compare to that of large language models in biomedical binary NLI classification tasks?**

   The TF-IDF baseline achieved a macro F1-score of 0.46, showing poor performance on both entailment and contradiction cases. All tested LLMs outperformed it significantly, with Qwen2 reaching 0.66 in zero-shot settings. This shows the limits of frequency-based similarity methods in capturing clinical semantics.

2. **Which prompt structures are most effective across different LLM architectures?**

   Prompts that integrated the hypothesis directly into the question consistently improved performance across all models (see Appendix for prompt format). Structural changes helped Qwen2 the most, likely due to its stronger instruction alignment. These prompt variations enabled better task understanding and more balanced predictions.

3. **What is the impact of few-shot learning on model performance, and which model architecture benefits most from few-shot prompts in this setting?**

Two-shot prompts improved performance for all models. Qwen2 showed the greatest gain, reaching 0.69 F1, while Flan-T5 and Mistral also benefited. Two-shot prompting provided clearer context and reduced output variance.

4. **Can lightweight fine-tuning methods improve the performance of the best-performing model?**

Qwen2 was selected for fine-tuning. Using LoRA, the model improved to 0.74 macro F1. This confirmed that domain adaptation with efficient methods can significantly improve output without full model retraining.

5. **How can we achieve stable and reliable predictions when applying inference models to biomedical NLI?**

Applying self-consistency voting after fine-tuning raised performance further to 0.78 F1. An odd voting count (5) ensured majority prediction. This approach stabilized borderline predictions and reduced inconsistencies across similar inputs.

5.4 Limitations

This study faced several constraints that limited the scope of model exploration and technique implementation.

- **Limited Computational Resources:**

Due to the absence of high-end GPUs, full fine-tuning of large models was not possible. The project relied on Low-Rank Adaptation (LoRA) instead. This limitation also restricted the evaluation of larger or proprietary models such as GPT-4 or Med-PaLM.

- **Restricted Dataset Scope:**

Only the NLI4CT dataset was used for training and evaluation. Although it provided high-quality annotations in a real-world biomedical setting, other biomedical NLI datasets were not included, either because they were not publicly available or required preprocessing efforts beyond the project timeline.

- **Limited Model Variety:**

The study focused on Flan-T5, Mistral, and Qwen2. Other competitive models, such as LLaMA 3, Claude, and Gemini, were excluded due to restricted access and hardware limitations.

- **No Use of Augmented CoT Reasoning:**

Augmented Chain-of-Thought (CoT) prompting was not implemented, despite its potential to improve reasoning in inference tasks. The added complexity and computational cost exceeded available resources.

5.5 Future Work

Several directions remain open to improve model robustness and generalization in biomedical entailment classification:

- **Structured Output Control:**

Incorporating step-by-step reasoning structures during inference, especially in CoT setups, may enhance prediction interpretability and reduce inconsistency in outputs.

- **Advanced Self-Consistency Techniques:**

Beyond majority voting, future work can explore weighted aggregation or confidence-based filtering strategies to stabilize model outputs more effectively.

- **Broader Evaluation Scope:**

Expanding the evaluation to include more diverse biomedical NLI datasets and larger-scale language models would offer better insight into model generalizability and performance boundaries across domains.

5.5 Conclusion

This thesis explored how different modeling choices affect performance in biomedical entailment classification using the NLI4CT dataset. The study began with a traditional TF-IDF approach and moved through several configurations of large language models; Flan-T5 XL, Mistral-7B, and Qwen2-7B-Instruct, under zero-shot, two-shot, and fine-tuned conditions.

The statistical baseline using TF-IDF showed limited ability to capture the complexity of clinical language. In contrast, transformer-based models performed more reliably. Among them, Qwen2-7B-Instruct emerged as the strongest option based on its performance in prompt-based evaluations.

This model was then fine-tuned using the LoRA method, which allowed targeted learning without retraining the full model. The fine-tuned version improved its macro F1-score from 0.66 to 0.74. To stabilize outputs further, a self-consistency method was added. By generating multiple outputs and selecting the most frequent label, the model's performance reached a macro F1-score of 0.78.

Compared to other systems submitted to SemEval-2024, which used ensembles and multi-step pipelines, the setup in this study achieved similar or better results using a single model and limited computational resources. These outcomes show that careful design of prompts, fine-

tuning techniques, and post-processing steps can produce strong results in domain-specific tasks without increasing system complexity.

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APPENDICES

Appendix A. Prompt Templates

A.1 Zero-Shot Prompt Format

```
Input:
Does the following statement logically follow from the clinical trial
description(s)?
Primary trial evidence are: {Primary Trial}
Secondary trial evidence are: {Secondary Trial}
Question: Does this imply that {Statement}?
Answer with only one of the options (Entailment or Contradiction):
Output: {label}
```

A.2 Two-Shot Prompt Format

```
Input:
Here are my first examples:
Example 1:
Primary trial evidence are: {example1_Primary Trial}
Question: Does this imply that {example1_Statement}?
Answer with only one of the options (Entailment or Contradiction): Entailment
Here are my second examples:
Example 2:
Primary trial evidence are: {example2_Primary Trial}
Question: Does this imply that {example2_Statement}?
Answer with only one of the options (Entailment or Contradiction):
Contradiction
Now, classify the following example using only 'ENTAILMENT' or
'CONTRADICTION':
Primary trial evidence are: {Primary Trial}
Question: Does this imply that {Statement}?
Answer with only one of the options (Entailment or Contradiction):
Output: {label}
```